RA-04: Clustered-gap Krylov bounds

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Status. The general assertion in RA-04 is not proved or disproved in this report. The results below establish the requested iteration order for exactly repeated b-sized leading clusters, for arbitrary admissible spectra with $t = \lceil k/b \rceil = 2$, for all admissible spectra with $\Delta \leq 1/k'$, and for several endpoint or exact-rank cases. Two weaker bounds are proved for every admissible input. The missing step for the unrestricted assertion is stated explicitly, rather than used as an assumption inside a purported proof.

Abstract. A useful object for clustered-gap analysis is the graph of a Krylov subspace over its leading eigenspace, not the smallest singular value of its raw monomial representation. Elementary interpolation and Gaussian small-ball estimates give quantitative graph bounds. A scalar filter recovers the general $k' - b + 1$ dependence. A truncated fractional-moment argument reduces it to $t \log(2k'/\Delta)$, proving the requested order when $\Delta \leq 1/k'$. Exact repeated clusters and arbitrary spectra with $t = 2$ admit the requested dependence for every positive clustered gap. A separate three-dimensional example rules out a scale-free lower bound for a normalized raw monomial Krylov matrix; this example does not refute RA-04. Exact symbolic checks and reproducible numerical experiments accompany the proofs.

Contents

1 Statement and scope

Let $A \in \mathbb{R}^{n\times d}$, put $M = AA^T$, and order the eigenvalues of M as $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Thus $\lambda_i = \sigma_i(A)^2$, with zeros appended when needed. Set

$$1 \leq b \leq k, \qquad t = \lceil k/b \rceil, \qquad m = bt = k' \leq \operatorname{rank}(A),$$

and define

$$\Delta = \min_{1\leq i\leq m-b} \frac{\lambda_i - \lambda_{i+b}}{\lambda_i} > 0. \tag{1}$$

For $m = b$ the minimum is empty and $\Delta = 1$. Let $G \in \mathbb{R}^{n\times b}$ have independent standard normal entries, and let

$$\mathcal{K}_q(M, G) = \operatorname{range}[\, G, MG, \ldots, M^{q-1}G\,]. \tag{2}$$

If Z is an orthonormal basis of this space, the output is $\widehat{A} = Z[Z^T A]_k$, where the bracket denotes an SVD truncation.

RA-04 asks whether a universal constant C makes

$$q = \left\lceil \frac{C}{\sqrt{\varepsilon}} \left[t \log \frac{2}{\Delta} + \log \frac{n}{\delta\varepsilon} \right] \right\rceil \tag{3}$$

sufficient, for $0 < \varepsilon, \delta < 1/2$, to guarantee probability at least $1-\delta$ for

$$\left\| A - \widehat{A} \right\|_\xi \leq (1+\varepsilon)\, \|A - [A]_k\|_\xi\,, \qquad \xi \in \{2, F\}, \tag{4}$$

$$\left| \|Av_i\|_2^2 - \lambda_i \right| \leq \varepsilon\lambda_{k+1}, \qquad 1 \leq i \leq k, \tag{5}$$

where the v_i are the ordered top k *right* singular vectors of $\widehat{A}$. These are the precise algorithm and guarantees considered here.[1] All mathematical results below concern exact arithmetic.

The distinction between what is proved and what remains is summarized below. Here and throughout, a constant in a restricted-case conclusion is uniform in all the parameters still allowed to vary in that case.

Regime or statement	Result in this report
Every input satisfying (1)	Two bounds: a larger $m-b+1$ term, or an extra $t\log m$ term; Theorems 4.1 and 5.3.
$\Delta \leq 1/m$, arbitrary b, t	The requested order; Corollary 5.4.
Exactly t leading levels, each repeated b times	The requested order (3); Theorem 6.1.
$t = 2$, with arbitrary leading eigenvalues	The requested order (3); Theorem 7.1.
$b = k$	A Gaussian starting-block proof; Corollary 6.2.
$\operatorname{rank}(A) = m$	Exact optimal approximation at $q \geq t+1$ almost surely; Proposition 8.1.
Arbitrary b, t and arbitrary clustered spectra	No proof of (3), and no counterexample, is established.

The general $m-b+1$ bound is a reconstruction of the earlier type of small-block interpolation argument, not a claim that this order is new. Meyer, Musco, and Musco analyze this regime in their Theorem 4.5 and Appendix E.[2] No novelty priority is asserted for the other results in this report either.

[1] Open Problems in Numerical Linear Algebra, RA-04, "Clustered singular-value gaps in randomized block Krylov approximation", accessed September 12, 2026. The entry distinguishes this algorithmic question from separate raw-matrix conditioning conjectures.

[2] R. A. Meyer, C. Musco, and C. Musco, *On the Unreasonable Effectiveness of Single Vector Krylov Methods for Low-Rank Approximation*, SODA 2024, pp. 811–845, Theorem 4.5 and Appendices E–F.

2 A basis-invariant reduction

Write an orthogonal eigendecomposition

$$M = U\operatorname{diag}(\Lambda, \Lambda_\perp)U^T, \qquad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_m).$$

By orthogonal invariance of a standard Gaussian matrix,

$$U^T G = \begin{bmatrix} H \\ T \end{bmatrix}, \qquad H \in \mathbb{R}^{m\times b}, \quad T \in \mathbb{R}^{(n-m)\times b},$$

again has independent standard normal entries. Work in these coordinates until returning to the original matrix at the end of an argument.

Definition 2.1 (Graph certificate)**.** A graph certificate at depth q_0 is a matrix

$$Y = U\begin{bmatrix} I_m \\ F \end{bmatrix} \quad \text{whose columns lie in } \mathcal{K}_{q_0}(M, G). \tag{6}$$

Its graph size is $L = 1 + \|F\|_2^2$.

Lemma 2.2 (Graph size controls leading-subspace overlap)**.** *If* (6) *holds, then for every* $1 \le j \le m$ *there is an orthonormal* $Q \in \mathbb{R}^{n\times j}$ *in the same Krylov space such that*

$$\left\|(U_j^T Q)^{-1}\right\|_2^2 \le 1 + \|F\|_2^2 .$$

Proof. Let E_j consist of the first j columns of I_m, put $F_j = FE_j$, and set

$$Q = YE_j(I_j + F_j^T F_j)^{-1/2}.$$

The identity $Y^T Y = I_m + F^T F$ gives $Q^T Q = I_j$. Moreover, $U_j^T Q = (I_j + F_j^T F_j)^{-1/2}$. Therefore

$$\left\|(U_j^T Q)^{-1}\right\|_2^2 = \left\|I_j + F_j^T F_j\right\|_2 \le 1 + \|F\|_2^2 .$$

This proof also handles a target rank k not divisible by b. □

For completeness, the one imported algorithmic result is stated explicitly. The property in the preceding lemma is usually called a (k, L)*-good starting block.*

Theorem 2.3 (Imported Krylov convergence theorem)**.** *There is a universal* C_0 *with the following property. Suppose the column space of a starting matrix* B *contains an orthonormal* $Q \in \mathbb{R}^{n\times k}$ *satisfying* $\|(U_k^T Q)^{-1}\|_2^2 \le L$*. If*

$$s \ge \max\left\{2, \left\lceil \frac{C_0}{\sqrt{\varepsilon}} \log \frac{nL}{\varepsilon} \right\rceil\right\},$$

then the projected SVD obtained from $\mathcal{K}_s(M, B)$ *has the approximation and right-component guarantees* (4)–(5).

This is the convergence transfer stated as Imported Theorem 3.2 in Chen et al.; their Problem 1.1 explicitly uses right singular vectors. Their theorem invokes the good-start analysis of Meyer, Musco, and Musco, Appendix F.[3] The proof of this imported convergence theorem is not rederived here. In particular, no

[3] T. Chen, E. N. Epperly, R. A. Meyer, C. Musco, and A. Rao, *Does block size matter in randomized block Krylov low-rank approximation?*, arXiv:2508.06486v2, Problem 1.1 and Imported Theorem 3.2; SODA 2026, pp. 1026–1046, doi:10.1137/1.9781611978971.42.

inference from a left-component guarantee alone is being silently substituted for the right-component condition.

If $B = [G, MG, \dots, M^{q_0-1}G]$, multiplication of the displayed powers gives

$$\mathcal{K}_s(M, B) = \mathcal{K}_{s+q_0-1}(M, G). \tag{7}$$

Consequently, a graph certificate with

$$\log L \le C_1 \left(t \log \frac{2}{\Delta} + \log \frac{n}{\delta} \right) \quad \text{and} \quad q_0 \le C_2 \left(t \log \frac{2}{\Delta} + \log \frac{n}{\delta} \right) \tag{8}$$

with probability $1 - \delta$ would prove RA-04. The logarithms absorb fixed integer offsets, since $\varepsilon, \delta < 1/2$ and $\log(2/\Delta) \ge \log 2$.

For a square simulated head block, define

$$K_H = [H, \Lambda H, \dots, \Lambda^{t-1} H], \qquad K_T = [T, \Lambda_\perp T, \dots, \Lambda_\perp^{t-1} T]. \tag{9}$$

When K_H is invertible the canonical graph is

$$F = K_T K_H^{-1}. \tag{10}$$

For every invertible coefficient-basis change R, this expression satisfies $(K_T R)(K_H R)^{-1} = K_T K_H^{-1}$. Thus it is unaffected by changing from monomials to a different basis of scalar polynomials. A smallest singular value of K_H alone has no such invariance.

3 Elementary probability and separation facts

Lemma 3.1 (Gaussian estimates)**.** *Let g be a standard real normal variable, let $X \in \mathbb{R}^{a\times b}$ be a standard Gaussian matrix, and let $S \in \mathbb{R}^{b\times b}$ be a standard Gaussian matrix. For $u, R > 0$,*

$$\mathbb{P}\{|g| \le u\} \le u, \tag{11}$$

$$\mathbb{E}\, \|X\|_F^2 = ab, \tag{12}$$

$$\mathbb{P}\{\|S^{-1}\|_2 > R\} \le \frac{b^{3/2}}{R}. \tag{13}$$

The inverse exists almost surely.

Proof. The density of g is at most $1/\sqrt{2\pi}$, so the probability of an interval of length $2u$ is at most $\sqrt{2/\pi}\, u \le u$. The second assertion follows by summing the unit second moments of all entries.

For the last assertion, let s_i^T be row i of S, and let d_i be its distance to the span of the other rows. The ith column of S^{-1} has norm d_i^{-1}: it is orthogonal to all the other rows and has inner product one with s_i. Hence

$$\|S^{-1}\|_2 \le \|S^{-1}\|_F \le \sqrt{b} \max_i d_i^{-1}.$$

Conditioned on the other rows, d_i has the distribution of $|g|$. Their span has dimension $b - 1$ almost surely, since a Gaussian matrix has full row rank almost surely. The union bound and (11) give

$$\mathbb{P}\{\|S^{-1}\|_2 > R\} \le \sum_{i=1}^{b} \mathbb{P}\{d_i < \sqrt{b}/R\} \le b^{3/2}/R.$$

The full-rank assertion follows, for example, because the determinant is a nonzero polynomial and a Gaussian law is absolutely continuous. □

Lemma 3.2 (A relative window contains at most b leading eigenvalues)**.** *For every $i \le m$, the interval*

$$I_i = [(1-\Delta/2)\lambda_i, (1+\Delta/2)\lambda_i] \tag{14}$$

contains at most b of $\lambda_1, \dots, \lambda_m$, counting multiplicities. Thus there is a deterministic set $J_i \subseteq \{1, \dots, m\}$ of size b containing i and all the indices in this window. For $j \notin J_i$,

$$\frac{\lambda_i}{|\lambda_j - \lambda_i|} \le \frac{2}{\Delta}, \qquad \frac{\lambda_j}{|\lambda_j - \lambda_i|} \le \frac{3}{\Delta}. \tag{15}$$

Proof. The assertion is immediate when $m = b$. Otherwise, $b+1$ eigenvalues in I_i would yield some $a \le m-b$ with

$$\frac{\lambda_{a+b}}{\lambda_a} \ge \frac{1-\Delta/2}{1+\Delta/2} > 1-\Delta,$$

contradicting (1). Fill the window indices to a set of size b by any fixed rule depending only on the eigenvalues.

For an excluded index, $|\lambda_j - \lambda_i| > (\Delta/2)\lambda_i$, proving the first inequality. If $\lambda_j > \lambda_i$, then

$$\frac{\lambda_j}{\lambda_j - \lambda_i} = 1 + \frac{\lambda_i}{\lambda_j - \lambda_i} \le 1 + \frac{2}{\Delta} \le \frac{3}{\Delta}.$$

If $\lambda_j < \lambda_i$, replace the numerator by λ_i and use the first bound. The same inequalities remain valid at $m = b$, when there are no excluded indices. □

Lemma 3.3 (The square head Krylov matrix is generically invertible)**.** *Under* (1)*, K_H in* (9) *is invertible almost surely.*

Proof. Its determinant is a polynomial in the entries of H. To show this polynomial is not identically zero, set row i of H equal to $e^T_{1+((i-1) \bmod b)}$. Permuting rows and columns makes K_H block diagonal, with one $t \times t$ scalar Vandermonde block for each residue class modulo b. The nodes in class r are $\lambda_r, \lambda_{r+b}, \dots, \lambda_{r+(t-1)b}$, and are strictly decreasing by (1). Every block is invertible. Absolute continuity of the Gaussian distribution finishes the proof. □

Generic invertibility is only an almost-sure existence statement. It does not give the uniform high-probability graph bound (8).

4 A general clustered-gap bound with a larger leading term

Theorem 4.1 (General scalar-filter bound)**.** *Put $d_0 = m - b$. For every admissible input, with probability at least $1-\delta$ there is a graph certificate at depth $q_0 = d_0 + 1$ with*

$$L \le 1 + \frac{8nbm^3}{\delta^3}\left(\frac{3}{\Delta}\right)^{2d_0}. \tag{16}$$

In particular, a universal constant C makes

$$q \ge \left\lceil \frac{C}{\sqrt{\varepsilon}} \left[(m-b+1)\log\frac{3}{\Delta} + \log\frac{n}{\delta\varepsilon}\right]\right\rceil \tag{17}$$

sufficient for (4)–(5).

Proof. For each $i \le m$, use J_i from Lemma 3.2. Let c_i be a unit vector orthogonal to the $b-1$ Gaussian rows h_j^T with $j \in J_i \setminus \{i\}$. Choose it measurably from those rows alone. When $b = 1$ take $c_i = 1$. The scalar

$$\alpha_i = h_i^T c_i$$

is standard normal conditioned on all rows other than i. Independence between different α_i is neither asserted nor needed.

Define the scalar polynomial

$$p_i(x) = \prod_{j \notin J_i} \frac{x - \lambda_j}{\lambda_i - \lambda_j}. \tag{18}$$

All its denominators are nonzero, and its degree is d_0. Let

$$y_i = p_i(M) G c_i / \alpha_i.$$

The ith leading coordinate of $U^T y_i$ equals one. A different leading coordinate is zero either because its index is in J_i and its Gaussian row annihilates c_i, or because its index is outside J_i and the polynomial vanishes. Thus $Y = [y_1, \ldots, y_m]$ has the form (6) and lies in $\mathcal{K}_{d_0+1}(M, G)$.

Every tail eigenvalue x belongs to $[0, \lambda_m]$. For $j \le m$, $|x - \lambda_j| \le \lambda_j$. Equation (15) therefore gives

$$\max_{0 \le x \le \lambda_m} |p_i(x)| \le \left(\frac{3}{\Delta}\right)^{d_0} =: \rho. \tag{19}$$

By the scalar small-ball bound and a union bound,

$$\mathbb{P}\left\{\min_i |\alpha_i| < \frac{\delta}{2m}\right\} \le \delta/2.$$

Markov's inequality gives

$$\mathbb{P}\left\{\|T\|_F^2 > \frac{2nb}{\delta}\right\} \le \delta/2.$$

On the intersection of the complementary events,

$$\begin{aligned} \|F\|_2^2 \le \|F\|_F^2 &= \sum_{i=1}^{m} \|p_i(\Lambda_\perp) T c_i / \alpha_i\|_2^2 \\ &\le m\rho^2 \, \|T\|_F^2 \left(\frac{2m}{\delta}\right)^2 \le \frac{8nbm^3}{\delta^3}\rho^2. \end{aligned}$$

This proves (16).

Because $b, m \le n$, the prefactor nbm^3 is at most n^5. Taking logarithms in (16), applying Lemma 2.2 and Theorem 2.3, and using (7) proves (17). □

This proof tolerates exact repetitions among the leading eigenvalues and requires no consecutive-gap or leading-condition-number assumption. Its limitation is the polynomial degree $m - b = b(t-1)$, which gives the wrong leading dependence when both b and t are allowed to grow. It does not establish the unrestricted target (3).

5 An all-input bound with one extra logarithmic factor

The scalar-filter construction pays for $b(t-1)$ individual zeros. A recursive vector-polynomial argument avoids this factor of b, at the cost of an extra $t \log m$ term. In particular, it proves the full requested order in the additional regime $\Delta \le 1/m$.

For this section write

$$E_{\lambda,H}(x) = [I_b, xI_b, \dots, x^{t-1}I_b]K_H^{-1}. \tag{20}$$

This is a $b \times m$ matrix polynomial. Its ith column is the vector polynomial with tangential values $h_j^T p_i(\lambda_j) = \mathbf{1}_{i=j}$.

Lemma 5.1 (Recursive leave-one-out identity)**.** *For $t \geq 2$, fix i, take J_i from Lemma 3.2, and put $S_i = \{1, \dots, m\} \setminus J_i$, of size $(t-1)b$. Almost surely there is a vector polynomial P_i of degree at most $t-1$, chosen from all rows other than i, such that*

$$h_j^T P_i(\lambda_j) = 0 \quad (j \neq i), \qquad z_i = P_i(\lambda_i), \qquad \|z_i\|_2 = 1.$$

Let $E_{S_i}(x)$ denote the interpolation evaluation matrix for the subproblem with nodes and Gaussian rows indexed by S_i, and put $D_i = \mathrm{diag}((\lambda_j - \lambda_i)^{-1} : j \in S_i)$. Then

$$P_i(x) = z_i - (x - \lambda_i)E_{S_i}(x)D_i H_{S_i} z_i, \tag{21}$$

so in particular

$$\|P_i(0)\|_2 \leq 1 + \frac{2}{\Delta}\left\|E_{S_i}(0)\right\|_2 \left\|H_{S_i}\right\|_F. \tag{22}$$

With $\alpha_i = h_i^T z_i$, column i of $E_{\lambda,H}(0)$ equals $P_i(0)/\alpha_i$, and α_i is standard normal conditioned on all other rows.

Proof. Deleting row i of the almost surely invertible K_H leaves a matrix of rank $m-1$. Choose a nonzero vector in its one-dimensional nullspace, measurably using only the other rows, and regard its t coefficient blocks as a vector polynomial P_i.

Any sorted subset of the head nodes inherits a b-step relative gap of at least Δ: if its indices are $j_1 < j_2 < \cdots$, then $j_{r+b} \geq j_r + b$, and the assertion follows from monotonicity of the original sequence. Thus the square $(t-1)$-step head matrix on S_i is invertible almost surely by Lemma 3.3.

If $P_i(\lambda_i) = 0$, write $P_i(x) = (x - \lambda_i)Q_i(x)$. At every $j \in S_i$ the factor $\lambda_j - \lambda_i$ is nonzero, so $h_j^T Q_i(\lambda_j) = 0$. The subproblem's invertibility would force $Q_i = 0$, a contradiction. Consequently $z_i \neq 0$, and P_i can be normalized as stated without using h_i.

Now write $P_i(x) = z_i + (x - \lambda_i)Q_i(x)$. For $j \in S_i$,

$$h_j^T Q_i(\lambda_j) = -\frac{h_j^T z_i}{\lambda_j - \lambda_i}.$$

Uniqueness of the degree-$(t-2)$ subproblem interpolant gives $Q_i(x) = -E_{S_i}(x)D_i H_{S_i} z_i$, proving (21). Since $\lambda_i \|D_i\|_2 \leq 2/\Delta$, (22) follows. The final assertions follow by conditioning on all other rows and using tangential interpolation uniqueness in the original problem. □

Theorem 5.2 (A truncated fractional-moment bound)**.** *Let $0 < \eta < 1/2$ and define*

$$R_\eta = \sqrt{2mb + 4\log(2/\eta)}, \qquad B_\eta = \frac{4}{\eta^2}(6m)^{2t}\left(\frac{2R_\eta}{\Delta}\right)^{t-1}. \tag{23}$$

Then

$$\mathbb{P}\{\left\|E_{\lambda,H}(0)\right\|_2 \leq B_\eta\} \geq 1 - \eta, \qquad \log B_\eta = O\left(t\log\frac{2m}{\Delta} + \log\frac{2}{\eta}\right), \tag{24}$$

with universal implied constants, for every admissible head spectrum.

Proof. For a standard normal scalar g, set $c_* = \mathbb{E}|g|^{-1/2} < 3$. Indeed the integral on $|g| \le 1$ is at most $4/\sqrt{2\pi} < 2$, and the integral on $|g| > 1$ is at most one.

Fix $R \ge 1$ and a dimension cap m. For an s-step subproblem of size $bs \le m$ with relative gap at least Δ, put

$$a_s(R) = \mathbb{E}\left[\|E(0)\|_2^{1/2}\,\mathbf{1}_{\{\|H\|_F \le R\}}\right].$$

For the recurrence, take the supremum of this expectation over all such spectra, so the same $a_s(R)$ bounds every subproblem. At $s = 1$, $E(0) = H^{-1}$. The norm of its ith column is the reciprocal of the absolute value of a standard normal scalar conditioned on all other rows. The inequalities

$$\|E(0)\|_2^{1/2} \le \|E(0)\|_F^{1/2} \le \sum_i \|E(0)e_i\|_2^{1/2}$$

therefore give $a_1(R) \le 3b \le 3m$.

For $s \ge 2$, use Lemma 5.1 for each column. Condition on every row except i. The indicator of $\|H\|_F \le R$ can first be bounded by the indicator of $\|H_{-i}\|_F \le R$ and then by that of $\left\|H_{S_i}\right\|_F \le R$. On this latter event, (22) and $(u+v)^{1/2} \le u^{1/2} + v^{1/2}$ give

$$\mathbb{E}\left[\|E(0)e_i\|_2^{1/2}\,\mathbf{1}_{\{\|H\|_F \le R\}}\right] \le 3\left[1 + \left(\frac{2R}{\Delta}\right)^{1/2} a_{s-1}(R)\right].$$

Here the subproblem rows remain an ordinary Gaussian matrix; no independence between its inverse and its entries is assumed. Summing over at most m columns proves the uniform recurrence

$$a_s(R) \le 3m\left[1 + \left(\frac{2R}{\Delta}\right)^{1/2} a_{s-1}(R)\right].$$

Induction, using $R \ge 1$ and $\Delta \le 1$, yields

$$a_s(R) \le (6m)^s \left(\frac{2R}{\Delta}\right)^{(s-1)/2}. \tag{25}$$

This truncation is important: it avoids asserting an independence that would be false, and it avoids multiplying the failure exponent by t.

For $X = \|H\|_F^2$, independence and the Gaussian integral give $\mathbb{E}e^{X/4} = 2^{mb/2}$. Consequently

$$\mathbb{P}\{X > 2mb + 4\log(2/\eta)\} \le 2^{mb/2} e^{-mb/2}\,\eta/2 \le \eta/2.$$

Apply Markov's inequality to the truncated half moment in (25) with $R = R_\eta$ and $s = t$:

$$\mathbb{P}\{\|E(0)\|_2 > B_\eta,\ \|H\|_F \le R_\eta\} \le \frac{(6m)^t (2R_\eta/\Delta)^{(t-1)/2}}{\sqrt{B_\eta}} = \eta/2.$$

Together these prove the probability bound.

For the logarithmic claim, let $x = \log(2/\eta)$. Since $b, t \le m$,

$$R_\eta \le 2m + 2\sqrt{x}, \qquad (t-1)\log R_\eta \le (t-1)\log(2m) + \sqrt{x}.$$

Taking logarithms in (23) and using $\sqrt{x} \le x + 1$ proves (24). □

Theorem 5.3 (All-input clustered-gap iteration bound)**.** *There is a universal constant C such that, for every input in RA-04,*

$$q \ge \left\lceil \frac{C}{\sqrt{\varepsilon}}\left[t\log\frac{2m}{\Delta} + \log\frac{n}{\delta\varepsilon}\right]\right\rceil \tag{26}$$

is sufficient for (4)–(5) *with probability at least* $1 - \delta$.

Proof. For $0 \le x < \lambda_m$, translation of the interpolation polynomial gives $E_{\lambda,H}(x) = E_{\lambda-x,H}(0)$. The shifted positive spectrum has b-step relative gaps at least Δ. Apply Theorem 5.2 at the t interior Chebyshev points

$$x_j = \frac{\lambda_m}{2}\left(1 + \cos\frac{(2j-1)\pi}{2t}\right)$$

with $\eta = \delta/(2t)$ for each point. A union bound and the elementary matrix-polynomial interpolation inequality proved in Lemma 10.2 give, with probability at least $1 - \delta/2$,

$$\sup_{0\le x\le\lambda_m} \left\|E_{\lambda,H}(x)\right\|_2 \le (2t-1)B_{\delta/(2t)}.$$

Independently of any possible dependence among these head events, $\|T\|_F^2 \le 2nb/\delta$ fails with probability at most $\delta/2$. The canonical graph at depth t therefore satisfies

$$L \le 1 + \frac{2nb}{\delta}(2t-1)^2 B^2_{\delta/(2t)} \tag{27}$$

with probability at least $1 - \delta$. Its logarithm is $O(t\log(2m/\Delta) + \log(n/\delta))$. Apply the graph reduction and Theorem 2.3. □

Corollary 5.4 (The requested order when $\Delta \le 1/m$)**.** *RA-04's requested iteration count is sufficient, with a universal constant, for all its inputs satisfying* $\Delta \le 1/m$.

Proof. In this regime $\log(2m/\Delta) \le 2\log(2/\Delta)$, so (26) implies (3) after increasing the universal constant. □

Theorem 5.3 is not the full requested bound: at a constant clustered gap it contains $t\log m$, rather than t alone. For unrestricted inputs one may use the better of (17) and (26); the choice depends only on the deterministic problem parameters. The two bounds need not be realized on the same Gaussian event.

6 The requested order for exact b-sized clusters

Theorem 6.1 (Exactly repeated leading clusters)**.** *Suppose*

$$\lambda_{(r-1)b+1} = \cdots = \lambda_{rb} = \mu_r, \qquad \mu_1 > \cdots > \mu_t > 0. \tag{28}$$

No additional hypothesis is imposed on the tail, including on repetitions of μ_t there. With probability at least $1-\delta$, a graph certificate exists at depth $q_0 = t$ with

$$L \le 1 + \frac{8nt^3b^4}{\delta^3}\Delta^{-2(t-1)}. \tag{29}$$

Consequently, (3) *is sufficient for* (4)–(5)*, with a universal constant.*

Proof. Partition H into t square row blocks $H_r \in \mathbb{R}^{b\times b}$. Use the scalar Lagrange polynomials

$$\ell_r(x) = \prod_{s\ne r}\frac{x-\mu_s}{\mu_r-\mu_s}.$$

Each has degree $t-1$, and $\ell_r(\mu_s) = \mathbf{1}_{r=s}$. Almost surely every H_r is invertible. The matrices

$$Y_r = \ell_r(M)GH_r^{-1}, \qquad Y = [Y_1,\ldots,Y_t]$$

therefore have leading block I_m, and their columns lie in $\mathcal{K}_t(M,G)$.

For $x \in [0, \mu_t]$, $|x - \mu_s| \le \mu_s$. If $s < r$, then $\mu_s - \mu_r \ge \mu_s - \mu_{s+1} \ge \Delta\mu_s$. If $s > r$, then $\mu_r - \mu_s \ge \mu_r - \mu_{r+1} \ge \Delta\mu_r \ge \Delta\mu_s$. Consequently,

$$|\ell_r(x)| \le \Delta^{-(t-1)} \quad (0 \le x \le \mu_t). \tag{30}$$

This also holds for $t = 1$ by the empty-product convention.

Lemma 3.1 and the union bound imply

$$\mathbb{P}\left\{\max_r \left\|H_r^{-1}\right\|_2 > \frac{2tb^{3/2}}{\delta}\right\} \le \delta/2.$$

Intersect this with $\|T\|_F^2 \le 2nb/\delta$, whose failure probability is at most $\delta/2$. The tail block is the horizontal concatenation of $\ell_r(\Lambda_\perp)TH_r^{-1}$, so

$$\begin{aligned}\|F\|_2^2 &\le \sum_{r=1}^{t} \left\|\ell_r(\Lambda_\perp)TH_r^{-1}\right\|_F^2 \\ &\le t\Delta^{-2(t-1)}\frac{2nb}{\delta}\left(\frac{2tb^{3/2}}{\delta}\right)^2,\end{aligned}$$

which is (29).

Since $t^3b^4 = bm^3 \le n^4$, the logarithm of this bound is $O(t\log(2/\Delta) + \log(n/\delta))$ with an absolute implied constant. The graph reduction proves the algorithmic conclusion. □

Corollary 6.2 (Endpoints)**.** *For $b = 1$, Theorem 6.1 applies to every admissible spectrum, since its clusters are singletons. For $b = k$ and $t = 1$, no equality among the leading eigenvalues is needed: the graph $GH^{-1} = U[I_b; TH^{-1}]$ gives* (29) *directly and proves the requested order.*

The exact-cluster argument cannot be extended to a nonzero-width cluster by simply assigning a common label to its eigenvalues. At distinct nodes, $\ell_r(\lambda_j)$ is no longer the required block indicator. That loss of an exact interpolation identity is a genuine missing step, not a small notational modification.

7 The requested order for arbitrary spectra when $t = 2$

The next result does not require exact repetitions or an a priori partition into separated clusters. The only spectral assumption is (1).

Theorem 7.1 (Two-step simulated block)**.** *Suppose $m = 2b$ and $\Delta > 0$. With probability at least $1 - \delta$, the canonical graph at depth $q_0 = 2$ satisfies*

$$L \le 1 + \frac{4000\,n^{12}}{\delta^6\Delta^2}. \tag{31}$$

Thus there is a universal constant C such that

$$q \ge \left\lceil \frac{C}{\sqrt{\varepsilon}}\left[\log\frac{2}{\Delta} + \log\frac{n}{\delta\varepsilon}\right]\right\rceil \tag{32}$$

gives all the guarantees in RA-04. In particular, (3) *holds whenever $\lceil k/b \rceil = 2$.*

Proof. Use the sets J_i from Lemma 3.2, and let $S_i = \{1, \ldots, m\} \setminus J_i$. Both sets have size b. Write H_{S_i} for the square Gaussian matrix consisting of those rows.

Delete row i from $K_H = [H, \Lambda H]$. Almost surely the remaining matrix has rank $m-1$, by Lemma 3.3. Its nullspace is represented by a nonzero vector-valued linear polynomial

$$P_i(x) = a_i + xc_i \in \mathbb{R}^b, \qquad h_j^T P_i(\lambda_j) = 0 \quad (j \neq i).$$

Choose this nullspace representative measurably using only the other rows. Let $z_i = P_i(\lambda_i)$.

First, $z_i \neq 0$ almost surely. Otherwise $a_i = -\lambda_i c_i$ and, for $j \in S_i$,

$$0 = (\lambda_j - \lambda_i) h_j^T c_i.$$

All these scalar factors are nonzero and H_{S_i} is invertible almost surely, so $c_i = 0$ and then $a_i = 0$, a contradiction. We may therefore normalize the chosen representative so that $\|z_i\|_2 = 1$.

For $j \in S_i$, the same nullspace equations now give

$$h_j^T c_i = -\frac{h_j^T z_i}{\lambda_j - \lambda_i}.$$

With

$$D_i = \operatorname{diag}\left(\frac{1}{\lambda_j - \lambda_i} : j \in S_i\right),$$

we obtain the exact identities

$$c_i = -H_{S_i}^{-1} D_i H_{S_i} z_i, \tag{33}$$

$$P_i(x) = \left[I_b - (x - \lambda_i) H_{S_i}^{-1} D_i H_{S_i}\right] z_i. \tag{34}$$

For $0 \leq x \leq \lambda_m$ we have $|x - \lambda_i| \leq \lambda_i$. Equation (15) therefore implies the uniform estimate

$$\|P_i(x)\|_2 \leq 1 + \frac{2}{\Delta} \left\|H_{S_i}^{-1}\right\|_2 \|H_{S_i}\|_2. \tag{35}$$

The scalar $\alpha_i = h_i^T z_i$ is standard normal conditional on the other rows, because z_i has unit norm and is independent of row i. The polynomial $P_i(x)/\alpha_i$ is the unique vector-valued polynomial whose head evaluations are the ith coordinate vector. Thus, at a tail node x with Gaussian row g^T, column i of the graph has value $g^T P_i(x)/\alpha_i$.

Consider the following three events, with M_0, R_0, a_0 as specified:

$$\|G\|_F^2 \leq M_0 := 3nb/\delta,$$
$$\max_i \left\|H_{S_i}^{-1}\right\|_2 \leq R_0 := 3mb^{3/2}/\delta,$$
$$\min_i |\alpha_i| \geq a_0 := \delta/(3m).$$

Markov's inequality, (13), and (11), respectively, bound their failure probabilities by $\delta/3$ each. The sets S_i are determined by the spectrum, not by H, so the square-Gaussian inverse estimate applies to every one of them. No independence between these three events is needed.

On their intersection, (35) gives

$$\sup_{i,x} \|P_i(x)\|_2 \leq B_0 := 1 + \frac{2R_0\sqrt{M_0}}{\Delta}.$$

Summing the squared column bounds for the tail matrix yields

$$\|F\|_2^2 \leq \|F\|_F^2 \leq mM_0 a_0^{-2} B_0^2 = \frac{27nbm^3}{\delta^3}\left(1 + \frac{6mb^{3/2}}{\delta\Delta}\sqrt{\frac{3nb}{\delta}}\right)^2. \tag{36}$$

To simplify, use $b, m \le n$, $\delta < 1$, and $\Delta \le 1$. Then

$$mM_0 a_0^{-2} \le 27 n^5 \delta^{-3}, \qquad B_0 \le 12 n^{7/2} \delta^{-3/2} \Delta^{-1}.$$

The product is at most $3888 n^{12} \delta^{-6} \Delta^{-2}$, proving (31) with the rounded constant 4000. Taking logarithms and applying the graph reduction proves (32). ◻

The essential feature of (33) is that a linear vector polynomial has only one unknown slope vector. After deleting a row, its slope is controlled by one square Gaussian submatrix at spectrally far indices. For degree $t - 1 > 1$, division by $x - \lambda_i$ leaves a vector polynomial of degree $t - 2$, rather than a single vector. The displayed argument then no longer closes with one square-Gaussian inverse bound.

8 Exact recovery when the rank equals k'

Proposition 8.1. *If* $\operatorname{rank}(A) = m = bt$ *and* (1) *holds, then at every* $q \ge t + 1$ *the output* $\widehat{A}$ *is a best rank-k approximation of A almost surely. Its ordered right singular vectors satisfy* $\|Av_i\|_2^2 = \lambda_i$ *for* $i \le k$.

Proof. In the eigenbasis, the tail of M is zero. Therefore

$$U^T [MG, M^2G, \ldots, M^tG] = \begin{bmatrix} \Lambda K_H \\ 0 \end{bmatrix}.$$

Both Λ and K_H are invertible almost surely. The columns in this display span $\operatorname{range}(A)$ and are contained in $\mathcal{K}_{t+1}(M, G)$. Consequently $ZZ^TA = A$. Multiplication by the isometry Z preserves the nonzero singular structure of Z^TA, so its SVD truncation is an optimal rank-k approximation of A, with the asserted right-component energies. ◻

This includes the zero-tail-error case $\operatorname{rank}(A) = k$. The assumptions then force $m = k$, hence b divides k. No division by $\lambda_{k+1} = 0$ is used in the argument.

9 A counterexample to a stronger raw-conditioning assertion

Proposition 9.1 (Raw monomial conditioning is not gap-only)**.** *Fix $b = 1$ and $m = t = 3$. There is a family with $\lambda_1 = 1$ and $\Delta = 1/2$ for which the smallest singular value of K_H converges to zero in probability. In particular, there is no strictly positive, spectrum-uniform lower bound for that singular value depending only on m, b, Δ, and a fixed failure probability.*

Proof. For $0 < \eta \le 1/2$, take

$$\Lambda_\eta = \operatorname{diag}(1, \eta, \eta/2), \qquad H = (g_1, g_2, g_3)^T.$$

Then

$$K_\eta = \operatorname{diag}(g_1, g_2, g_3) \begin{bmatrix} 1 & 1 & 1 \\ 1 & \eta & \eta^2 \\ 1 & \eta/2 & \eta^2/4 \end{bmatrix}, \qquad \Delta = \min\{1 - \eta, 1/2\} = 1/2.$$

For the unit vector $c = (0, -1, 1)^T/\sqrt{2}$,

$$K_\eta c = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ g_2(\eta^2 - \eta) \\ g_3(\eta^2/4 - \eta/2) \end{bmatrix}.$$

Consequently

$$\sigma_{\min}(K_\eta) \le \frac{\eta}{\sqrt{2}}\sqrt{g_2^2 + g_3^2/4}, \qquad \mathbb{E}\sigma_{\min}(K_\eta)^2 \le \frac{5}{8}\eta^2. \tag{37}$$

For every fixed $a > 0$, Markov's inequality gives $\mathbb{P}\{\sigma_{\min}(K_\eta) \ge a\} \le 5\eta^2/(8a^2) \to 0$. $\square$

This is not a counterexample to RA-04. It concerns the coordinates of a particular monomial basis. The algorithm uses its span. In fact, $b = 1$ is covered by Theorem 6.1, and the scalar Lagrange tail bounds there remain uniform for this family. The proposition explains why discarding a condition-number factor from a raw-matrix lower bound is not, by itself, a valid route to the algorithmic result.

10 The remaining quantitative question

Recall the interpolation evaluation matrix $E_{\lambda,H}(x)$ in (20). Its ith column is the vector polynomial whose tangential head evaluations are $h_j^T p_i(\lambda_j) = \mathbf{1}_{j=i}$. At a tail node x_j, with Gaussian row g_j^T, the canonical graph row is $g_j^T E_{\lambda,H}(x_j)$.

Proposition 10.1 (A sufficient interpolation estimate)**.** *Suppose universal constants $a, c > 0$ existed such that, for every admissible head spectrum and every $0 < \eta < 1/2$,*

$$\mathbb{P}_H\left\{\sup_{0\le x\le\lambda_m} \left\|E_{\lambda,H}(x)\right\|_2 \le \left(\frac{cm}{\eta}\right)^a \left(\frac{2}{\Delta}\right)^{ct}\right\} \ge 1-\eta. \tag{IE}$$

Then RA-04 would follow.

Proof. Apply (IE) with $\eta = \delta/2$ and intersect its event with $\|T\|_F^2 \le 2nb/\delta$. The latter has failure probability at most $\delta/2$. Row by row,

$$\|F\|_F^2 \le \|T\|_F^2 \sup_{0\le x\le\lambda_m} \left\|E_{\lambda,H}(x)\right\|_2^2.$$

The logarithm of the right-hand side is $O(t\log(2/\Delta) + \log(n/\delta))$. The graph has depth $q_0 = t$, so Theorem 2.3 and (7) give the claimed order. $\square$

Estimate (IE) is *not proved here for arbitrary b, t.* It is a sufficient condition, not an equivalent reformulation of RA-04: the algorithm may benefit from additional Krylov powers even if a square-depth interpolation estimate is poor.

There is a useful reduction of the uniform-in-x issue to a single extrapolation point.

Lemma 10.2 (Zero evaluation suffices up to polynomial factors)**.** *Suppose a bound of the form on the right-hand side of* (IE) *holds only for $\left\|E_{\lambda,H}(0)\right\|_2$, uniformly over positive head spectra. Then* (IE) *follows after changing the universal constants.*

Proof. A common shift of the nodes corresponds to translating a vector polynomial. Uniqueness of interpolation therefore gives

$$E_{\lambda,H}(x) = E_{\lambda-x,H}(0) \qquad (0 \le x < \lambda_m).$$

The shifted spectrum is positive and its b-step relative gaps satisfy

$$\frac{(\lambda_i - x) - (\lambda_{i+b} - x)}{\lambda_i - x} = \frac{\lambda_i - \lambda_{i+b}}{\lambda_i - x} \ge \Delta.$$

Apply the asserted zero-evaluation bound at the t points

$$x_j = \frac{\lambda_m}{2}\left(1 + \cos\frac{(2j-1)\pi}{2t}\right), \qquad 1 \le j \le t,$$

with failure probability η/t for each. A union bound controls all these evaluations by $(cmt/\eta)^a(2/\Delta)^{ct}$.

For any matrix polynomial P of degree at most $t-1$ on $[-1,1]$, its Chebyshev expansion $P(y) = \sum_{r=0}^{t-1} C_r T_r(y)$ has coefficients

$$C_0 = \frac{1}{t}\sum_j P(y_j), \qquad C_r = \frac{2}{t}\sum_j P(y_j)\cos(r\theta_j) \quad (r \ge 1),$$

where $y_j = \cos\theta_j$ and $\theta_j = (2j-1)\pi/(2t)$. These identities follow from discrete cosine orthogonality. If $N = \max_j \left\|P(y_j)\right\|_2$, then $\|C_0\|_2 \le N$ and $\|C_r\|_2 \le 2N$, whence

$$\sup_{-1\le y\le 1} \|P(y)\|_2 \le (2t-1)N.$$

Apply this to $P(y) = E_{\lambda,H}(\lambda_m(1+y)/2)$. Since $t \le m$, the extra factor $(2t-1)t^a$ is polynomial in m and is absorbed into new universal exponents in (IE). □

10.1 Why several tempting replacements do not prove the estimate

A scalar root count alone is insufficient. A degree-$t-1$ scalar polynomial can be small near $t-1$ roots, and each such neighborhood may contain b leading eigenvalues. Thus as many as $b(t-1)$ head rows can be affected. Passing from a root count to a uniform Gaussian subspace estimate is exactly the quantitative issue.

Nor can a general jointly Gaussian smallest-singular-value theorem be applied without verifying its hypotheses. For example, Nie's relevant anti-concentration theorem assumes a nontrivial small-ball estimate for *every* unit bilinear form.[4] A diagonal Gaussian matrix is invertible almost surely but has $e_1^T Z e_2 = 0$ identically. Exact-cluster reductions can produce just such block-decoupled structures. Almost-sure invertibility is not a verification of that stronger premise.

Finally, scalar Lagrange products do not carry over by simply replacing nodes with noncommuting matrices. In the right-coefficient convention,

$$\Phi(xI) = (xI - B_2)(xI - B_3) = x^2 I - x(B_2 + B_3) + B_2 B_3$$

satisfies $\Phi(B_2) = 0$ but

$$\Phi(B_3) = B_2 B_3 - B_3 B_2.$$

For

$$B_2 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \qquad B_3 = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix},$$

this last matrix is $\left[\begin{smallmatrix} 0 & -1 \\ 0 & 0 \end{smallmatrix}\right]$, not zero, despite disjoint real spectra. A matrix-polynomial representation is therefore not by itself a norm bound for the needed interpolants. Shao's structural analysis is a relevant formulation of these matrix-interpolation issues, with remaining random quantities stated conjecturally.[5]

[4] Z. Nie, *Matrix anti-concentration inequalities with applications*, arXiv:2111.05553v2, Theorem 1.6; STOC 2022.

[5] N. Shao, *A structural bound for cluster robustness of randomized small-block Lanczos*, arXiv:2507.10144v2, May 30, 2026, Sections 2–3. The version was inspected through its PDF because the available HTML rendering omitted mathematical expressions.

11 Reproducible checks and their limits

The accompanying scripts separate exact identity checks from floating-point experiments. Their results are stored as JSON and CSV, with seeds and parameters. None of these finite tests is a substitute for (IE) or a proof of the unrestricted assertion.

11.1 Exact rational and symbolic identities

The script `tests/exact_checks.py` uses SymPy rational arithmetic. All seven test groups passed. They verify: an explicit residue-class Vandermonde full-rank witness; the exact-cluster graph and its Krylov membership; the general scalar-filter graph and its head identity; all six leave-one-out identities in a $b = 3, t = 2$ fixture; 23 general recursive identities across three further rational fixtures; the raw-conditioning test-vector identity; and the noncommuting product counterexample.

For example, the generic witness has determinant $-1{,}382{,}400$. The exact-cluster fixture has graph tail Frobenius norm squared 7,811,469,433/288,000,000. These rational fixtures are deliberately not described as Gaussian samples. The probability arguments remain the analytic proofs in the preceding sections.

11.2 Numerical graph and approximation checks

The second script performs 50 $t = 2$ tests, with $b \in \{1, 2, 4, 8, 16\}$, and checks (35) at 21 tail points for every interpolation column. The largest observed ratio of a column norm to its deterministic bound was 0.513813. Twelve additional single-tail graph probes used 120-decimal-digit arithmetic. Their relative linear-solve residuals were between approximately 10^{-128} and $3 \cdot 10^{-125}$; these are backward residuals, not certified forward-error bounds.

The randomized block Krylov experiment uses two-pass reorthogonalization and numerical-rank detection in binary64 arithmetic. It records 117 combinations of case, seed, and depth, directly computing all three requested error measures, including the energies of the *right* singular vectors. For $\varepsilon = 0.1$, the following is the smallest *tested* depth at which all three selected seeds satisfied all three nonzero-tail criteria. It is not a high-probability theorem or an estimate of the universal constant.

Case	n	b	k	m	Tested depth
Exact b-fold clusters	160	4	24	24	16
Near b-fold clusters	160	4	24	24	16
$t = 2$, unaligned spectrum	128	8	14	16	6
Nondivisible target rank	96	4	15	16	10
Mixed multiplicities	96	3	12	12	10
Scalar block	60	1	6	6	16

The exact-rank fixture provides an explicit reminder about arithmetic. For $n = 80$, $b = 4$, and $k = m = 24$, the mathematical theorem gives exact recovery at $q = 7$. In binary64, the three measured spectral residuals at $q = 7$ were about $4.09 \cdot 10^{-13}$, $2.48 \cdot 10^{-10}$, and $2.51 \cdot 10^{-12}$. The middle seed passed the script's 10^{-10} absolute tolerance only at $q = 8$. A finite-precision discrepancy is therefore retained in the results, not hidden as an exact-arithmetic success.

The raw-conditioning experiment uses 100-decimal-digit arithmetic and fixed values $(g_1, g_2, g_3) = (0.7, -1.3, 0.5)$. At $\eta = 10^{-4}, 10^{-12}, 10^{-24}$, the measured smallest singular values were approximately $1.650 \cdot 10^{-5}, 1.650 \cdot 10^{-13}, 1.650 \cdot 10^{-25}$, respectively, while Δ remained $1/2$. Proposition 9.1, rather than this finite sequence, establishes the limiting counterexample.

11.3 Reproduction

From the package root, run

```
python -m pip install -r requirements.txt
python tests/exact_checks.py --output results/exact_checks.json
python tests/numerical_checks.py --output-dir results
```

The scripts require Python 3.11 or later. Recorded versions are NumPy 2.3.5, SymPy 1.14.0, and mpmath 1.3.0; the supplied environment used Python 3.13.5. Results may differ slightly across floating-point libraries. The LaTeX source is `src/report.tex`; `build.sh` recompiles the PDF when a suitable TeX installation is available.

12 Conclusion

The graph bounds (29) and (31) yield the requested iteration order in their stated regimes. The all-input bound (26) retains only an extra $t \log m$ term, while (17) is sometimes sharper. Corollary 5.4 proves the requested order whenever $\Delta \leq 1/m$. The raw-conditioning example invalidates one stronger auxiliary route without invalidating the algorithmic assertion.

For arbitrary growing b and t with nonzero-width leading clusters and $\Delta > 1/m$, the required quantitative control has not been obtained. The sufficient interpolation estimate (IE), or a different argument exploiting additional Krylov powers, remains an unproved obligation in this report. Accordingly, the package is a collection of proved partial results and checks, not a complete solution of RA-04.

Sources

[1] Open Problems in Numerical Linear Algebra. *RA-04: Clustered singular-value gaps in randomized block Krylov approximation*. Repository entry, accessed September 12, 2026. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/randomized-and-low-rank-approximation/RA-04`. Used for the precise target, rank padding, and right-singular-vector convention.

[2] T. Chen, E. N. Epperly, R. A. Meyer, C. Musco, and A. Rao. *Does block size matter in randomized block Krylov low-rank approximation?* SODA 2026, pp. 1026–1046. arXiv:2508.06486v2. `https://arxiv.org/abs/2508.06486`; `https://doi.org/10.1137/1.9781611978971.42`. Used for the explicitly imported good-start convergence transfer.

[3] R. A. Meyer, C. Musco, and C. Musco. *On the Unreasonable Effectiveness of Single Vector Krylov Methods for Low-Rank Approximation*. SODA 2024, pp. 811–845. arXiv:2305.02535v3. `https://arxiv.org/abs/2305.02535`. Used for the relation to earlier small-block bounds and the convergence-transfer provenance.

[4] N. Shao. *A structural bound for cluster robustness of randomized small-block Lanczos*. arXiv:2507.10144v2, May 30, 2026. `https://arxiv.org/abs/2507.10144`. Used for the comparison with matrix-polynomial interpolation and its unresolved random factors.

[5] Z. Nie. *Matrix anti-concentration inequalities with applications*. STOC 2022; arXiv:2111.05553v2. `https://arxiv.org/abs/2111.05553`. Used only to identify the hypothesis that an attempted small-ball argument must verify.